

Lab Exercise 1

Write a program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros). Your program ends with the input 0. Display the average as a floating-point number. Use the class name **UnspecifiedNumbers**.

Here are some sample runs:

```
Enter integers ending with 0: 4 2 -1 3 -2 0
The number of positives is 3
The number of negatives is 2
The total is 6
The average is 1.2
```

```
Enter integers ending with 0: 0
no numbers are entered except 0
```

Lab Exercise 2

Write a program that prompts the user to enter the number of students and each student's name and score (at least two students), and finally displays the student with the highest score and the student with the second-highest score. Use the class name **TopStudents**.

Here are sample runs:

```
Enter the number of students: 5
Enter a student name: Mohammed
Enter a student score: 75
Enter a student name: Ali
Enter a student score: 85
Enter a student name: Fahad
Enter a student score: 98
Enter a student name: Khalid
Enter a student score: 65
Enter a student name: Saad
Enter a student score: 89
Top two students:
Fahad's score is 98.0
Saad's score is 89.0
```

```
Enter the number of students: 1
Need at least two students.
```

Lab Exercise 3

Write a Java program that calculates and prints the cost of games that a customer buys at a gaming store as following:

- The cost of the game is input.
- A customer must buy at least 1 game (otherwise print "Error").
- If a customer buys more than 2 games, then he will get a 20% discount.

The program should work as follows:

- It prints the following menu:

```
*****
*   Welcome to the Gaming Center :)   *
*   -----                           *
* 1) add ==> to add a game to inventory *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program      *
*****
```

- If the user typed "**add**":
 - Ask the user to type the number of games to add to the inventory.
 - Keep asking until the user types -1 as an answer.
- If the user typed "**sell**":
 - If there are no games in the inventory, print an error then show the menu.
 - Otherwise,
 - Ask the user if they want to add a game ("y" for yes, anything else for no).
 - If yes, ask the user to enter the game price.
 - Repeat until the user doesn't enter "y" for adding a game.
 - Stop the user from entering if there are no more games in the inventory.
 - Finally, show the receipt to the user. It must show "price before discount", "discount", and "price after discount".

Name your class **GameStore**.

Here is a sample run to show different cases:

```
*****
*   Welcome to the Gaming Center :)   *
*   -----                           *
* 1) add ==> to add a game to inventory *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program      *
*****
Enter your option :> sell
Sorry. There are no more games in store :(
```

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program          *
*****
```

```
Enter your option :> add
Please, number of games (-1 to end): 2
Please, number of games (-1 to end): 3
Please, number of games (-1 to end): -1
```

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program          *
*****
```

```
Enter your option :> sell
Do you want to add a game? ('y' for yes): y
Please, enter the price of next game: 100
Do you want to add a game? ('y' for yes): y
Please, enter the price of next game: 200
Do you want to add a game? ('y' for yes): y
Please, enter the price of next game: 120
Do you want to add a game? ('y' for yes): n
Total price before discount: 420.0SR
Your discount is: 84.0SR
Total price after discount: 336.0SR
```

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program          *
*****
```

```
Enter your option :> sell
Do you want to add a game? ('y' for yes): y
Please, enter the price of next game: 150
Do you want to add a game? ('y' for yes): no
Total price before the discount: 150.0SR
Your discount is: 0.0SR
Total price after discount: 150.0SR
```

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer *
* 3) exit ==> to end this program          *
*****
```

```
Enter your option :> sell
Do you want to add a game? ('y' for yes): y
Please, enter the price of next game: 100
Cannot sell more games. Out of stock :(
```

Total price before the discount: 100.0SR
Your discount is: 0.0SR
Total price after discount: 100.0SR

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer  *
* 3) exit ==> to end this program          *
*****
```

Enter your option :> play

Incorrect option

```
*****
*      Welcome to the Gaming Center :)      *
*      -----                              *
* 1) add ==> to add a game to inventory    *
* 2) sell ==> to sell games to a customer  *
* 3) exit ==> to end this program          *
*****
```

Enter your option :> exit

Thanks. Goodbye!